# JAWAHARLAL NEHRU UNIVERSITY

## SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

**Minor Exam**

**M.CA. (Winter Semester-2025)**

**Course Name: Computer Architecture** | **Maximum Marks: 25**

**Course Code:** CS 405 | **Total Time: 50 minutes**

Q1. **a)** What is the difference between RISC and CISC type of Computer Architecture?.

b) What is the role of different buses in a Computer system. Explain using 8085 microprocessor.

(3+2)

Q2. (a) What are the major performance parameters that determines the execution time of processor?.

b) Explain which performance parameters effects CPI and why?.

(a) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) scale upCPI instruction type C to 2.4. Which one is better?.

| Type | **CPI** | **Frequency** |
|---|---|---|
| A | 1.4 | 30% |
| B | 2.5 | 40% |
| C | 2.1 | 30% |

(4+2+4)

**Q3. (a)** What are the different types of machine cycle in 8085 microprocessor?.

(b) Write an assembly language program to subtract two 4 bit numbers. Determine the total T-state and calculate the total time consumed using 8085 microporocessor. What is the new memory capacity if the RAM memory is scaled upto from $2^{20}$ byte to $2^{34}$ byte?.

(4+4+2)

# JAWAHARLAL NEHRU UNIVERSITY

## SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

### Mid-Term Exam

### M.CA. (Winter Semester-2025)

**Course Name: Computer Architecture**
**Course Code:** CS 106

**Maximum Marks: 40**
**Total Time: 2 hour**

Q1. a) What is the difference between Computer Organization and Computer Architecture? Which parameters are directly related to processing power of the computer?.

b) Amdhal's law is applicable to parallel processing or pipeline processing?.

c) Write an assembly language program to show the difference between RISC and CISC Instruction set.

(4+2+4)

Q2. (a) What are different types of addressing modes in 8086 microprocessor

(b) A weather forecasting computation requires 40 Trillion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

(c) A scientific simulation program has the following execution profile:

- 50% of the workload is fully parallelizable.
- 30% of the workload is partially parallelizable across at most 8 processors.
- The remaining 20% is completely serial.

You upgrade the system to a 32-processor machine. Compute the program's overall speedup on the 32-processor machine. Compute the utilization of the 32 processors (i.e., the fraction of total potential processing power used).

(3+3+4)

Q3. (a) Explain the important parameters that defines the performance of a Memory System?. How can one increase the relative size in a Memory system?. What is the 90/10 rule?. What is the difference between Temporal and Spatial locality in memory system?.

b) Consider a CPU with an average CPI as 2.0. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles (=$t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(5+5)

Q4. (a) What are the total external connections in 256X32 Memory system?

(b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred is preferred and why?

(c) Explain the Arithmetic pipeline for performing multiplication. Deduce the reservation Table and determine its MAL?

(2+3+5)

$\frac{20}{40} \times 100$

# JAWAHARLAL NEHRU UNIVERSITY

## SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

### Mid-Term Exam

### M.CA. (Winter Semester-2025)

**Course Name: Computer Architecture**

**Course Code:** CS 106

**Maximum Marks: 40**

**Total Time: 2 hour**

Q1. a) Explain the working of SRAM cell

b) Explain RISC is better architecture for Parallel computing or Pipeline computing?

c) Write an assembly language program to show RISC is better than CISC Instruction set.

(4+2+4)

Q2. (a) Explain the additional addressing modes in 8086 microprocessors w.r.t. 8085 microprocessor.

(b) Explain Stack pointer and Program counter. (Use Push and Pop statement wherever necessary).Write an assembly language which results in the sum of first 9 Fibonacci numbers.

c) A scientific simulation program has the following execution profile:

- 35 % of the workload is fully parallelizable.
- 35 % of the workload is partially parallelizable across at most 12 processors.
- The remaining 30% is completely serial.

You upgrade the system to a 16-processor machine. Compute the program's overall speedup on the 16-processor machine. Compute the utilization of the 16 processors (i.e., the fraction of total potential processing power used). (2+5+3)

Q3. (a) How to improve the latency in a Memory System? Which level cache would be better for Temporal and Spatial locality in memory system and why?

b) Consider a CPU with an average CPI as 2.3. Assume the instruction mix is ALU-25%, Load-20%, Store-25% Branch-30%. Assume the cache miss rate of 10% and miss penalty of 60 cycles (=$t_{MM}$). Calculate the effective CPI for a combined L1 and L2 cache system.

(4+6)

Q4. (a) What is the difference between non linear dynamic and static pipelining technique?

(b) Which of the cache mapping is the best and worst and why?

© Explain the Arithmetic pipeline for performing subtraction. Deduce the reservation Table and determine its MAL?

(2+3+5)

# JAWAHARLAL NEHRU UNIVERSITY

## SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

**Final Exam**

**M.CA. (Winter Semester-2025)**

**Course Name: Computer Architecture**

**Course Code:** CS 106

**Maximum Marks: 40**

**Total Time: 3 hour**

Q1. a) Implement Universal gates using transistors.

b) What is the role of decoder and explain its role in a Computer system

c) What are the additional addressing modes in 8086 micro-processor. Write an assembly language program in 8086 to perform an XOR operation. `(3+2+5)

Q2. (a) Compute the total cycles required using the Harvard and Von Neumann Architecture where a loop adds 10 numbers stored in an array and stores the sum in memory. Instructions per iteration has Load number, add to accumulator and store accumulator. The Memory access is one cycle per fetch/store and total iterations are 10.

b) A program has 1 million instructions to be executed using either CISC processor whose CPI = 1.1, Clock = 2.0 GHz or RISC processor whose CPI = 2, Clock = 1.5 GHz. Which one is faster?.

c) Write an assembly language program to show the difference between Reduced Instruction set and Complex Instruction set.

(4+3+3)

Q3. (a) Consider a CPU with an average CPI as 2.5. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 15% and miss penalty of 120 cycles (=$t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(b) A computing task that can be divided into three parts:

1. Part A: 40% of the task can be fully parallelized.
2. Part B: 45% of the task can be parallelized.
3. Part C: 15% of the task is inherently serial and cannot be parallelized.

Calculate the overall speedup when using 12 processors for Part A and 4 processors for Part B. Part C remains serial.

(5+5)

Q4. (a) Explain the working of one unit of SRAM cell?

(b) What parameters play a pivitol role in deciding the memory hierarchy. What is the role of cache and virtual memory?

c) A computer has the following cache memory system: Cache size = 16 KB, Block size = 64 bytes, 4-way set associative cache, Main memory address = 16 bits. How many sets are in the cache?. How many bits are used for block offset, set index, and tag?

(3+3+4)

Q5. (a) What is the difference between static and dynamic pipelining techniques?

(b) Explain the different types of pipelining hazards?

© ) What is the difference between linear and non-linear pipelining.
b) For the following reservation table
(i) What are the forbidden latencies
(ii) Show the state transition diagram
(iii) List all the simple and greedy cycles
(iv) Determine the optimumal constant lattice cycle and MAL
(v)Determine the pipeline throughput with τ=15ns

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| S1 | X | | X | | | X |
| S2 | | X | X | X | | |
| S3 | X | | | | X | |

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| S1 | X | | | X | X |
| S2 | X | | X | | |
| S3 | | X | | X | |

(2+3+5)

Q6. (a) Write is a Vector processor?.

(b) Explain the difference between Tightly coupled and Loosely coupled microprocessor. Which one is preferred is preferred and why?
(c) What are the advantages and disadvantages of hardwired vs microprogrammed control units?

(2+4+4)